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SULTAN IDRIS EDUCATION UNIVERSITY

UNIVERSITI PENDIDIKAN SULTAN IDRIS

FINAL EXAMINATION
SEMESTER 1 SESSION 2018/2019

CODE: MES3103 **COURSE : SOFTWARE VERIFICATION AND VALIDATION**

DATE: 05 JAN 2019 **DURATION : 1 HOURS 30 MINUTES**

INSTRUCTIONS

1. This exam paper consists of **TWO (2)** parts only, namely **Part A** and **Part B**.
2. **Part A**: Answer **ALL** questions in the spaces provided.
3. **Part B**: Answer **ALL** questions in the spaces provided.

This question paper consists of 8 printed pages including front page

PROGRAMME : _____

YEAR : _____ CLASS GROUP : A

REGISTRATION NO : _____ | | | | | | |

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LECTURER : ASMA HANEE BINTI ARIFFIN

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PART A: (50 marks)

Answer **all** questions in the spaces provided.

1. Give brief definition for the following terminologies:
   (a) Software Testing
   (b) Fault
   (c) Bug
   (d) Error
   (e) Failure

[10 marks]

1 a)

1 b)

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1 c)

1 d)

1 e)

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2. Give brief comparisons for the following terminology pairs:
   (a) Verification versus Validation
   (b) Dynamic Testing versus Static Testing
   (c) Software Quality Assurance (SQA) Group versus Tester
   (d) Correctness versus Completeness
   (e) Verification by Inspection versus Verification by Analysis

[20 marks]

2 a)

2 b)

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2 c)

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2 d)

2 e)

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3. Software Quality Assurance (SQA) presents two important approaches that are used to determine the quality of the software, which are Defect Management Approach and Quality Attributes Approach.
   (a) List all **six (6)** quality characteristics in the Quality Attributes Approach. **[6 marks]**
   (b) Briefly describe **two (2)** of the quality characteristics listed in (a). **[4 marks]**

3 a)

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3 b)

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4. The reasons that make software review an important element of software development process are numerous. It is a methodology that offers an opportunity to the development team and the client to get clarity on the project as well as its requirements.
   (a) Give the definition of software peer review.
   [2 marks]
   (b) List **four (4)** types of software peer review.
   [4 marks]
   (c) Briefly describe **two (2)** types of software peer review listed in (b).
   [4 marks]

4 a)

4 b)

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4 c)

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PART B: (10 marks)

Answer **all** questions in the spaces provided.

Loan Process

Requirement specification Users want to apply for a bank loan in which there are different UI for different types of users.

1 Functional specification

1.1 New users
New users can access the bank URL and when they click on the "ApplyLoan" button, a page is displayed for them to enter their registration information.

A new user needs to provide first name, last name, address, DOB, SSN and email address mandatorily. If the user chooses to register the user will be asked to choose a password.

1.2 Existing users
Existing users can be the ones who have applied for a loan and are waiting on confirmation, or the ones that already have the loan granted.

The users who are at the waiting stage will have their information of in which stage the loan is displayed.
The loan, as soon as being applied, is in the stage of "Sent for review". Once it is reviewed, the stage moves to "Reviewed and approved" or "Reviewed and deleted" based on the banker's decision.

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2 Authentication

The different actors of this information system will not have the same rights and privileges. Authentication will follow the login/password pattern.

3 Ease of use

All the information should be accessible in less than 3 clicks for a user.

Figure 1 – Requirements and functional specifications for Loan Process.

1. Given the requirements and functional specifications in Figure 1, suggest any **five (5)** different ways of how we can verify and validate them.
   (Make sure that:
   (i) you clearly state whether it is a verification or validation for each of the way, and
   (ii) there must be a mixture of verification and validation in the whole suggestion.)

[10 marks]

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END OF QUESTIONS

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